



COMPOSITION NOTEBOOK

Models of Computation

Syntax of lambda

The calculus of lambda (λ -calculus) is an essential mathematical language that lies at the foundation of programming languages.

- Var (set of variables): its the finite set of the variables, i.e.: $x, y, z, \dots$;
- Λ (set of lambda-terms): its the set of all lambda terms, that is all valid expression that we can build.

We can define Λ with this formula:

$$\Lambda ::= \text{Var} \mid \Lambda \Lambda \mid (\lambda \text{Var}. \Lambda)$$

if CAN BE COMPOSED OF

So:

- Var:

$$\frac{m \in \text{Var}}{m \in \Lambda}$$

if m is a variable ($m \in \text{Var}$), THEN m is a valid term ($m \in \Lambda$)

- Application:

$$\frac{M \in \Lambda \quad N \in \Lambda}{(M N) \in \Lambda}$$

if M is a valid term ($M \in \Lambda$) and N too ($N \in \Lambda$), then we can write them next to each other ($M N \in \Lambda$)

- Abstraction:

$$\frac{m \in \text{Var} \quad M \in \Lambda}{(\lambda m. M) \in \Lambda}$$

if a variable $m \in \text{Var}$ and M is a valid term, then we can write $\lambda m. M$ and we obtain a new valid term

Exercise 1:

$$f(a, b) = a + b$$

We change the terms to make them more recognizable.

$$f(m, y) = m + y$$

We replace m with 3 and y with 7.

$$f(3, 7) = 3 + 7 = 10$$

λ can accept only one variable at a time. We can't write $\lambda(m, y)$, so we apply Abstraction (ABS)

2 times, one for m and one for y .

$$\frac{y \in \text{Var} \quad (m + y) \in \Lambda}{(\lambda y. m + y) \in \Lambda}$$

Subsequently, we apply it to m :

$$\frac{m \in \text{Var} \quad (\lambda y. m + y) \in \Lambda}{\lambda m (\lambda y. m + y) \in \Lambda}$$

So: $\lambda m (\lambda y. m + y)$

$$f = \lambda m. (\lambda y. m+y) \Rightarrow (\lambda m y. m+y)$$

Now, if we pass a number to the function, e.g. 3

$$(\lambda m y. m+y) 3$$

Subsequently, λm comes off because its mission is concluded

$$\lambda y. 3+y$$

We can stop here because with the next number i.e. 7 in this case, the sum will be equal to 10!

Exercise 2:

REMEMBER! THE PARENTHESES ARE HIDDEN. THE TRUE FORM IS: $\lambda m (\lambda y (\lambda z. m z (y z)))$

$$S = \lambda m y z. m z (y z)$$

Looking all the way to the left, we would have λm , so this is the first we delete

$$\boxed{\lambda m} . \boxed{(\lambda y z. m z (y z))}$$

m y z

using the abs rule $\frac{m \in \text{Var} \quad M \in \Lambda}{(\lambda m. M) \in \Lambda}$ bottom up, so:

... $\in \text{Var} / \Lambda$ WE CAN ALSO NOT WRITE IT EVERYWHERE

$$m \in \text{Var} \quad \lambda y z. m z (y z) \in \Lambda$$

Now we apply abs to the y :

$$y \in \text{Var} \quad \lambda z. m z (y z)$$

and now the same for z :

$$z \in \text{Var} \quad m z (y z)$$

Now we apply the Application rule (app) to the rule $m z$ and $y z$:

$$\boxed{m z} \quad \boxed{(y z)}$$

m z y z

$m z \in \Lambda$ $y z \in \Lambda$

We re-apply app to $m z$ and $y z$:

$$\boxed{m} \quad \boxed{z} \quad \boxed{y} \quad \boxed{z}$$

$m \in \Lambda$ $z \in \Lambda$ $y \in \Lambda$ $z \in \Lambda$

In the end we apply the Var rule to all 4 variables:

$$\boxed{m} \quad \boxed{z} \quad \boxed{y} \quad \boxed{z}$$

$m \in \text{Var}$ $z \in \text{Var}$ $y \in \text{Var}$ $z \in \text{Var}$

Calculus of Lambda: α -rule, β -reductions

α -RULE: the name we give to an "internal" variable in the function doesn't change the sense of the same function's meaning

β -REDUCTION: it's a lambda calculus rule and it applies a function to the value (substitution)

$$(\lambda m. M) N \xrightarrow{\beta\text{-REDUCTION}} M[N/m] \rightarrow \text{REDUCT}$$

Exercise 3:

$$S = (\lambda yz. m z (y z)) (\lambda a b. a) (\lambda c d. c)$$

STEP 1: β -reduction

$(\lambda a b. a)$ is an input value, being the first variable m disappears and is replaced:

$$(\lambda y z. (\lambda a b. a) z (y z)) (\lambda c d. c)$$

STEP 2: 2° β -reduction

we do the same with y and $(\lambda c d. c)$

$$\lambda z. (\lambda a b. a) z ((\lambda c d. c) z)$$

STEP 3.1: β -reduction

we apply the rule to the a
with z in input

$$\lambda b. z$$

STEP 3.2: β -reduction

we apply the rule to the c
with z in input

$$\lambda d. z$$

So:

$$\lambda z. (\lambda b. z) (\lambda d. z)$$

Passo 5: β -reduction

now we apply the rule to the b with $(\lambda d. z)$ in input

$$\lambda z. z$$

BEING THAT b DOESN'T EXIST TO THE RIGHT OF THE DOT, I CAN'T REPLACE ANYTHING